GHit!

Optimized Movie Concept Generator (OMCG) · Solution Design

Hit metric, judge model and LLM optimisation

1. Background

Making a film takes huge capital, labour and time. After years of production, a film can flop, turn out generic, or become a hit, and hits are rare. In our analysis, the hit we define as an AAA film happens only about 1 in 70 films: of 7,959 films with budget and revenue data, 39% earned 2.5× their budget and 15% were well liked, but only 1.4% did both at the top of their release year. Greenlight decisions still rest mostly on comparable titles and instinct, which don't tell a team what to make next.

This project builds a proof-of-concept (PoC) movie concept generator that helps production teams and writers get a step closer to an AAA-tier film.

2. Objectives

Business objectives

1. Help production teams and writers develop film concepts with a high chance of becoming AAA hits.
2. Increase the film's popularity.
3. Increase the film's revenue potential.

Analytical objectives

We frame this project as a highly imbalanced classification task and a text-generation task. Both first need a metric that defines an AAA film, which we approach as a clustering problem. The end goal is a system that generates film concepts (story text, genres, budget, runtime, director and two leads) that maximise the probability of becoming an AAA hit. Specifically:

1. Establish a statistically sound and business-meaningful definition of an AAA hit.
2. Train a judge (classifier) that estimates whether a given concept would be an AAA hit.
3. Optimise the prompt that drives the generator LLM, using the judge's scores.

3. Related studies

The papers below inspire the methodology and experimental setup. Together they show an LLM used not only to generate text but as an optimiser, paired with an automatic evaluator. Foundation models are also reaching tabular classification: TabPFN (Hollmann et al., Nature 2025) is reported to match or beat tuned gradient-boosted trees such as XGBoost and LightGBM on small tables. It costs more compute and latency, so we keep it as an optional judge candidate next to gradient boosting and logistic regression.

Paper	Idea	What we use
[1] Large Language Models as Optimizers (OPRO). Yang et al., ICLR 2024	The LLM is the optimiser. Each step, its meta-prompt lists past solutions with their scores, sorted low to high, and asks for better ones. 8 candidates per step, temperature 1.0.	The meta-prompt design (blocks ③ and ④ in Fig. 4) and the sampling settings

Paper	Idea	What we use
[2] Mathematical discoveries from program search with LLMs (FunSearch). Romera-Paredes et al., Nature 625, 2024	An LLM paired with an automatic evaluator, plus a database of scored solutions split into islands that evolve separately to stay diverse.	The judge as evaluator, and three genre islands. We skip island resets.
[3] Conservative Objective Models for Offline Model-Based Optimization. Trabucco, Kumar, Geng & Levine, ICML 2021	When you optimise against a model learned from fixed data, the search drifts to where the model is wrong.	The trust region: novelty and support limits, and a halved score off-distribution

4. Problem framing and modelling paradigm

The starting question is which concept *c* maximises P(AAA | c). We never observe that probability, only a judge $\hat{f}$ trained on past films. So the task is constrained, offline, black-box optimisation with an LLM as the proposal distribution:

$$\max_{c \sim p_{\mathrm{LLM}}(\cdot \mid \mathrm{prompt}_t)} \hat{f}(c) \quad \text{s.t.} \quad \max_j \cos(e_c, e_j) \le \tau_{\mathrm{nov}}, \;\; \mathrm{OOD}(c) \le \tau_{\mathrm{ood}}, \;\; c \text{ feasible}$$

Symbols and variables

Symbol	Meaning
c	A movie concept: story text (logline + synopsis), genres, budget, runtime, director and two leads.
$P(\mathrm{AAA} \mid c)$	The true probability that concept c becomes an AAA hit. Unknown for a film not yet made.
$\hat{f}(c)$	The judge's estimate of that probability (the hat means "estimated"): the median of $f_1, \ldots, f_{20}$, 20 calibrated refits on 80% of past films.
$p_{\mathrm{LLM}}(\cdot \mid \mathrm{prompt}_t)$	The distribution of concepts the generator LLM writes when given the prompt for round t. $c \sim p_{\mathrm{LLM}}$ means each concept is sampled from the LLM.
$\mathrm{prompt}_t,\ t$	The meta-prompt at round t = 1…5: task and constraints, evidence, the scored history so far, rejections.
s.t.	"Subject to": the conditions every concept must meet.
$e_c,\ e_j$	Embedding vectors of the plot of concept c and of existing film j; j runs over the ~44,000 films with a plot.
$\cos(e_c, e_j)$	Cosine similarity of the two plots: 1 = same direction (near-identical), 0 = unrelated. $\max_j$ picks the closest existing film.
τ_{nov}	Novelty threshold: the 99th percentile of film-to-nearest-film plot similarity (≈ 0.80). Above it, the concept counts as a copy and is rejected.
$\mathrm{OOD}(c)$	Out-of-distribution score: how unusual the concept's genre mix, budget, runtime and story are, as a percentile of past AAA-scale films (budget ≥ $100M), using the mean distance to the 10 nearest past films.
τ_{ood}	Support threshold: the 99th percentile. Above it, the concept is flagged UNFAMILIAR, its score is halved in the search and it can't enter the portfolio.
feasible	Director and leads from the shortlists (25 directors, 40 actors), budget $100–350M, runtime 90–175 min, original story, 1–3 allowed genres.

1. **Concepts, not tokens.** Optimising raw text would find wording that fools the judge. We generate concepts (the components of a film), not arbitrary combinations of tokens, and the LLM keeps every candidate a plausible, castable film.
2. **The objective is learned from biased data.** Only films that got made, 1.4% positives. The constraints keep the search where the judge has evidence [3].

3. **Association, not cause.** The judge ranks concepts; it can't promise outcomes. The output is a portfolio of three, one per island, for human review.

Component	Paradigm	Model
AAA label	Unsupervised ensemble → weak labels	KMeans, median split, DBSCAN, HDBSCAN (GLOSH), Isolation Forest; majority vote
Judge	Supervised binary classification, highly imbalanced (1.4% positives)	Gradient boosting (chosen over logistic regression; TabPFN optional, not run here), class-balanced, Platt-calibrated
Generator	In-context optimisation, no weight updates	GPT-6 Sol with an OPRO meta-prompt, FunSearch islands

5. High-level methodology

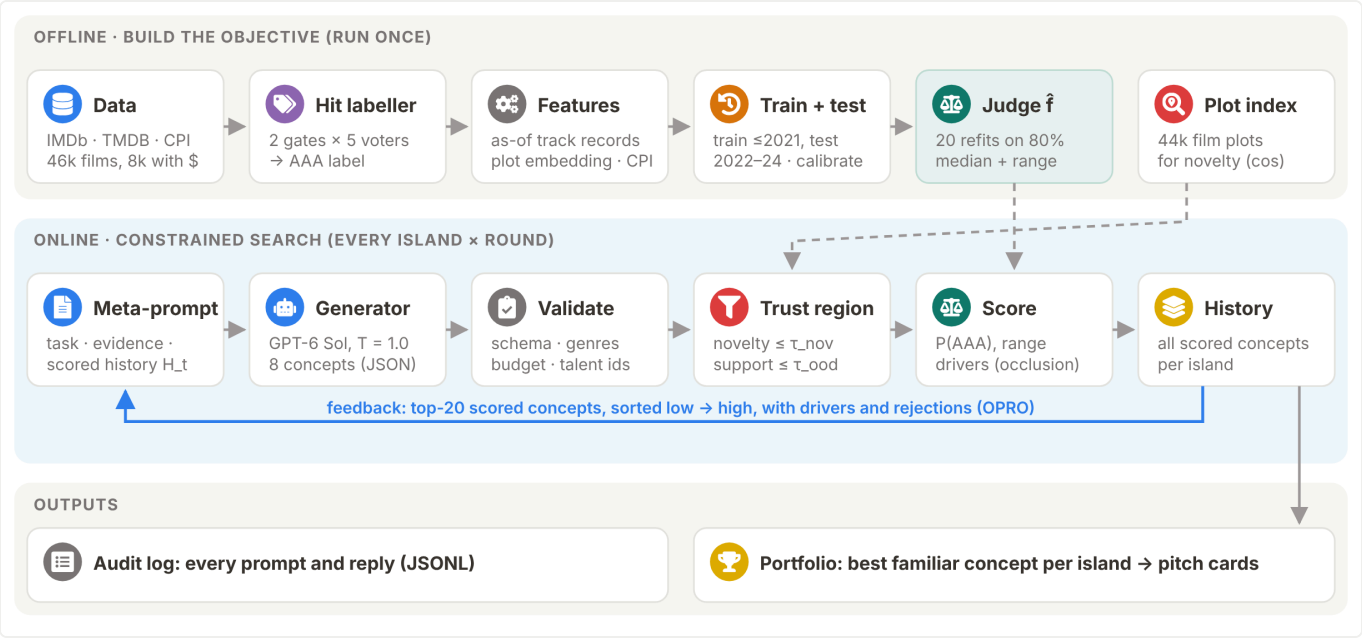


Fig. 1. Offline, we build the label, the judge and a plot index once. Online, each island runs 5 rounds of generate → check → score, and the scored history becomes the next prompt.

6. Data and features

Source	Used for	Rows
IMDb datasets	ratings, votes, genres, runtime, cast and crew	46,129 films
TMDB (v11)	budget, revenue, plot, keywords	7,959 with budget and revenue
US CPI (FRED)	2023 dollars	1970–2025

Judge features (pre-release only)

1. Genre mix: one-hot genres, number of genres
2. Budget (log, 2023 $), runtime, English flag
3. Source: original, sequel or adaptation
4. Director track record as of release: films, audience hits, AAA hits, smoothed rates
5. Lead cast track record, top two billed, same rule
6. Story: 16 principal components of the plot embedding (TF-IDF + SVD in this run)

Leakage controls: no votes, ratings or revenue of the film being scored; track records use earlier years only; release year excluded.

7. Step 1 · A statistically sound AAA label

Fig. 2 shows how the AAA label is built.

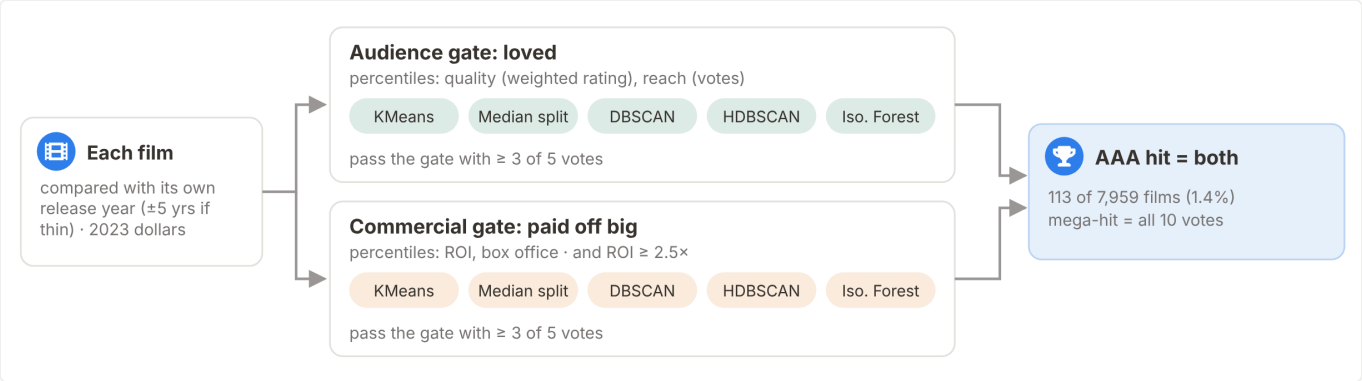


Fig. 2. Each gate is scored on within-year percentiles, so a 1985 film competes with 1985 films. Five methods vote, so no single clustering choice decides the label.

Gate	Share of 7,959 films	Films
Earned ≥ 2.5× its budget	39.2%	3,122
Loved (audience gate)	15.1%	1,205
Paid off big (commercial gate)	2.5%	202
Both gates: AAA hit	**1.4%**	**113**

- **Result:** 113 AAA hits, 1,092 loved but not top earners, 89 top earners nobody loved, 6,665 neither.
- **Agreement (Cohen's κ):** the methods form two camps. KMeans with the median split (κ 0.49 audience, 0.77 commercial) and HDBSCAN with Isolation Forest (κ 0.89, 0.77). Across camps κ ≤ 0.31, so a 3-of-5 vote needs both camps, which keeps the label strict.
- **Sanity check:** the all-time top three are Spider-Man: No Way Home (2021), The Lord of the Rings: The Two Towers (2002), Back to the Future (1985).

8. Step 2 · The judge

1. **Split by time.** Train on films up to 2018, pick the model on 2019–21, refit up to 2021, test once on 2022–24 (229 films, 5 AAA hits).
2. **Metric.** PR-AUC and its lift over the 2.2% base rate, since hits are rare. Brier score and a reliability plot for calibration.
3. **For concepts.** 20 refits on 80% subsamples; we report the median and the 10th–90th percentile.

Model (test 2022–24)	PR-AUC	Lift	ROC-AUC	Brier
Logistic regression	0.168	7.7×	0.926	0.0201
Gradient boosting (chosen)	0.167	7.6×	0.799	0.0205
Random	0.022	1.0×	0.5	–

- Both models rank AAA hits about 7.6× better than chance. Among the judge's top-ranked films, 1 in 3 were AAA hits, against 1 in 46 overall.

- Plot text and the director's track record carry most of the signal (Fig. 3a).
- With 5 test hits, read the lift as directional and the probabilities as ranking scores.

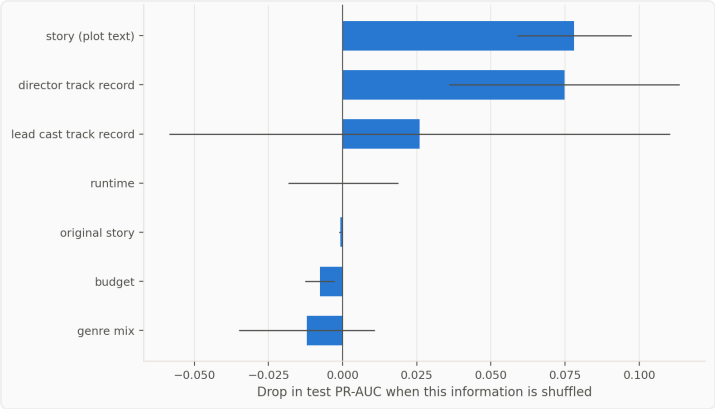


Fig. 3a. Drop in test PR-AUC when each feature group is shuffled.

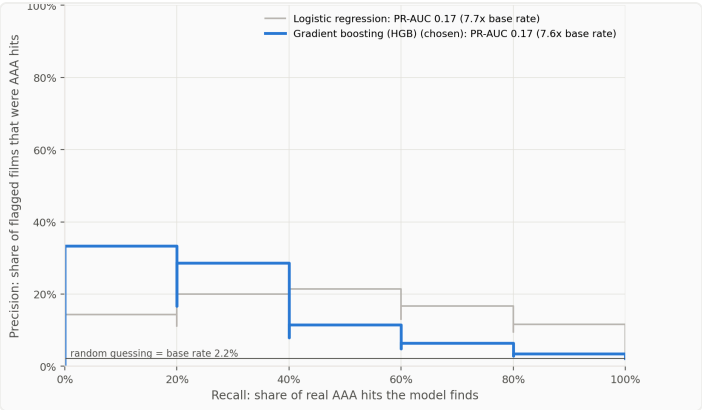


Fig. 3b. Precision–recall on 2022–24 releases. Grey line = random (2.2%).

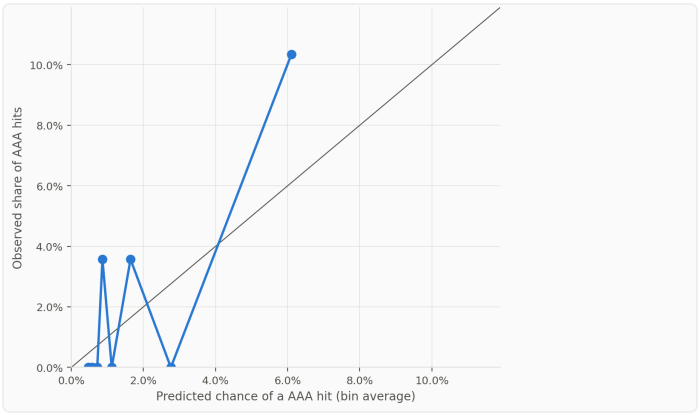


Fig. 3c. Reliability on 2022–24 (8 bins of ~29 films). The top bin predicted ~6% and saw 10%; with 5 hits the lower bins are noise.

9. Step 3 · Optimising the generator: OPRO × FunSearch

The LLM itself is the optimiser, and what changes from round to round is its prompt. After every round, the judge's scores are written back into the prompt, so the generator can see which ideas scored well and why.

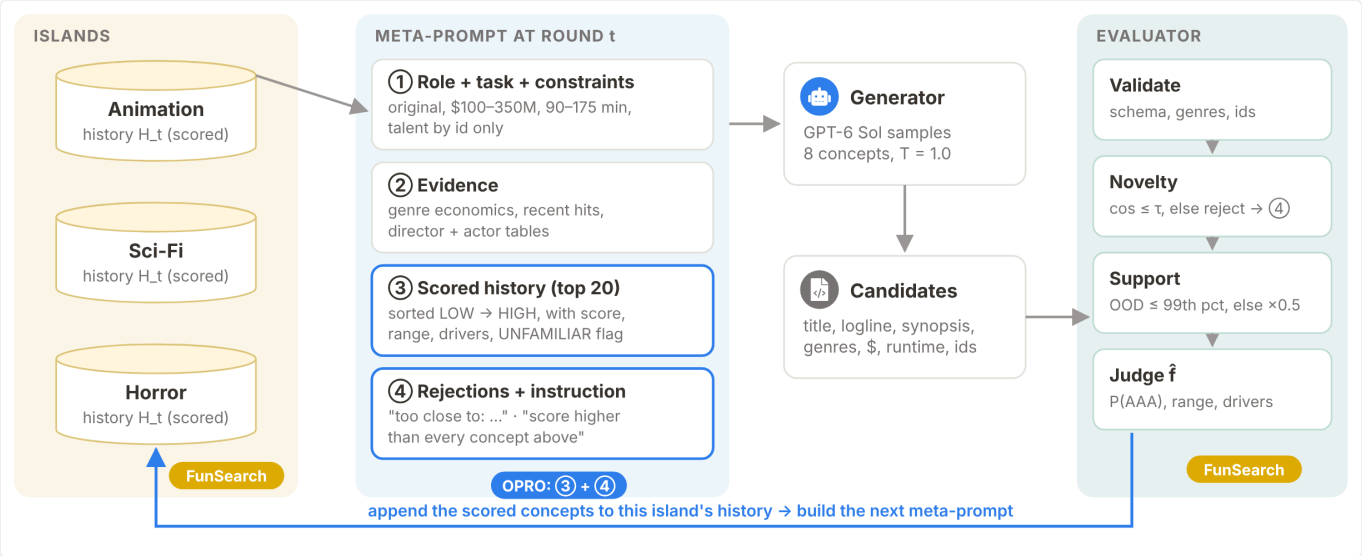


Fig. 4. One island, one round. The control arm makes the same number of calls with blocks ③ and ④ removed, so any gain can be traced to the feedback.

What the optimiser does each round

1. **Build the meta-prompt** for the island: task and constraints, evidence, then the 20 best concepts so far, sorted low to high (best last, closest to where the model writes), each with score, range, drivers and an UNFAMILIAR flag. Off-distribution concepts are ranked on half their score. Concepts rejected as copies are listed separately.
2. **Sample** 8 new concepts from GPT-6 Sol at temperature 1.0, as JSON.
3. **Evaluate** each one: schema and talent ids, novelty against 44,000 plots and earlier concepts, support, then the judge.
4. **Store** the scored concepts in the island's history and repeat for 5 rounds. Each island's best familiar concept goes to the portfolio.

Sample traces

① Round 1, generator output (scored)

```
8 concepts, e.g.
"The Dustline Express"  → 4.8%  (best)
"Bridge the Sky"        → 4.1%
"The Midnight Current"  → 1.5%  (worst)   driver: − story (−1.6 pts)
```

② Judge output for one concept

```
{"title": "Wildlife Hotline", "status": "scored",
 "p_aaa_hit": 0.065, "p_lo": 0.057, "p_hi": 0.287,
 "ood_pct": 47, "objective": 0.065,
 "drivers": [["director track record", 0.046],
             ["story (plot text)", 0.026],
             ["genre mix", 0.013]],
 "nearest_film": "Over the Hedge", "nearest_sim": 0.55}
```

③ What the optimiser sees next round (end of block ③, round 4)

```
... (lowest first)
[5.1%] "Zero Gravity Day" ... [UNFAMILIAR: unlike any past film, score unreliable]
[5.5%] "The Great Feast": Animation/Adventure/Comedy | $130M | dir D13 ...
[5.8%] "Lost and Howl": Animation/Adventure/Comedy | $125M | dir D02 ...
[6.2%] "Rain Check": Animation/Adventure/Family | $135M | 103 min | dir D01 | leads A05, A39
    drivers: + director track record (+4.8 pts), + story (plot text) (+2.4 pts), + genre mix (+0.9
pts)
Keep what the best concepts share; change what the worst share.
```

Island 1 (Animation), feedback arm	Round 1	Round 2	Round 3	Round 4	Round 5
Best P(AAA) in the round	4.8%	6.2%	5.8%	**6.5%**	5.9%
Mean P(AAA) in the round	3.5%	5.1%	4.8%	5.2%	4.7%
Best concept	The Dustline Express	Rain Check	Lost and Howl	**Wildlife Hotline**	Orbit Sweepers

The generator picks up what the judge rewards. In round 1, 62% of concepts were adventure or family films; from round 2 on, all of them were. The average director's smoothed AAA-hit rate rose from 23% to 42% in round 2. Both are what the top of the history had in common.

10. Results of the search

⚠️ **Pre-rescore.** These scores come from the run before the uncertainty-range fix. Rescoring the same 237 concepts will update every score and range, and may reorder the portfolio.

Concepts scored	Range of P(AAA)	Feedback vs control	Best vs base rate
237 from 30 calls; none too close to an existing film	**0.9%–6.5%**	**+0.3 pts** mean P(AAA) (p = 0.07)	**3.0×** the 2.2% base rate

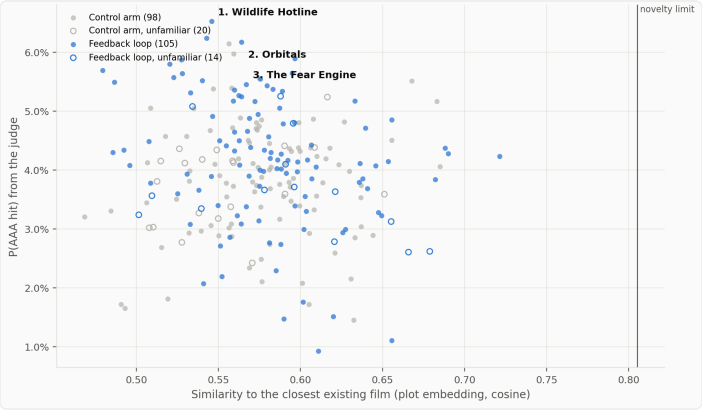


Fig. 5. Every concept: similarity to the closest film vs P(AAA). Right of the line would be a copy.

	Feedback	Control
Mean P(AAA)	4.10%	3.81%
Last-round mean	4.18%	3.53%

	Feedback	Control
Best per island (mean)	6.05%	5.83%
Within support	88%	83%

Paired by call (15 pairs): +0.29 pts, 95% CI [+0.03, +0.59], sign-flip p = 0.07. Feedback helps, but this run is too small to prove it.

The portfolio

#1 Wildlife Hotline · P(AAA) **6.5%** (range 5.7%–28.7%)

> *A rookie raccoon answering an animal-run emergency line must guide hundreds of stranded creatures across a sleeping city before their last habitat is sealed off.*
> Animation · Adventure · Comedy · $115M · 107 min · dir. Jonathan del Val · Ginnifer Goodwin, Chris Pratt

#2 Orbitals · P(AAA) **6.2%** (range 5.4%–24.6%)

> *When a solar storm disables Earth's navigation network, two obsolete satellites become the only guides for a spacecraft full of children.*
> Sci-Fi · Animation · Adventure · $145M · 103 min · dir. Kyle Balda · Chris Pratt, Idina Menzel

#3 The Fear Engine · P(AAA) **5.5%** (range 16.4%–42.4%)

> *A theme park's new ride reads guests' fears, then turns them into creatures throughout the resort; its designer must guide her son through attractions that know them both.*
> Horror · Adventure · Action · $145M · 121 min · dir. David F. Sandberg · Bryce Dallas Howard, Jaeden Martell

Range = 10th–90th percentile across judge refits. Full pitch cards, with synopsis, comparables and "why now", are in the deck.